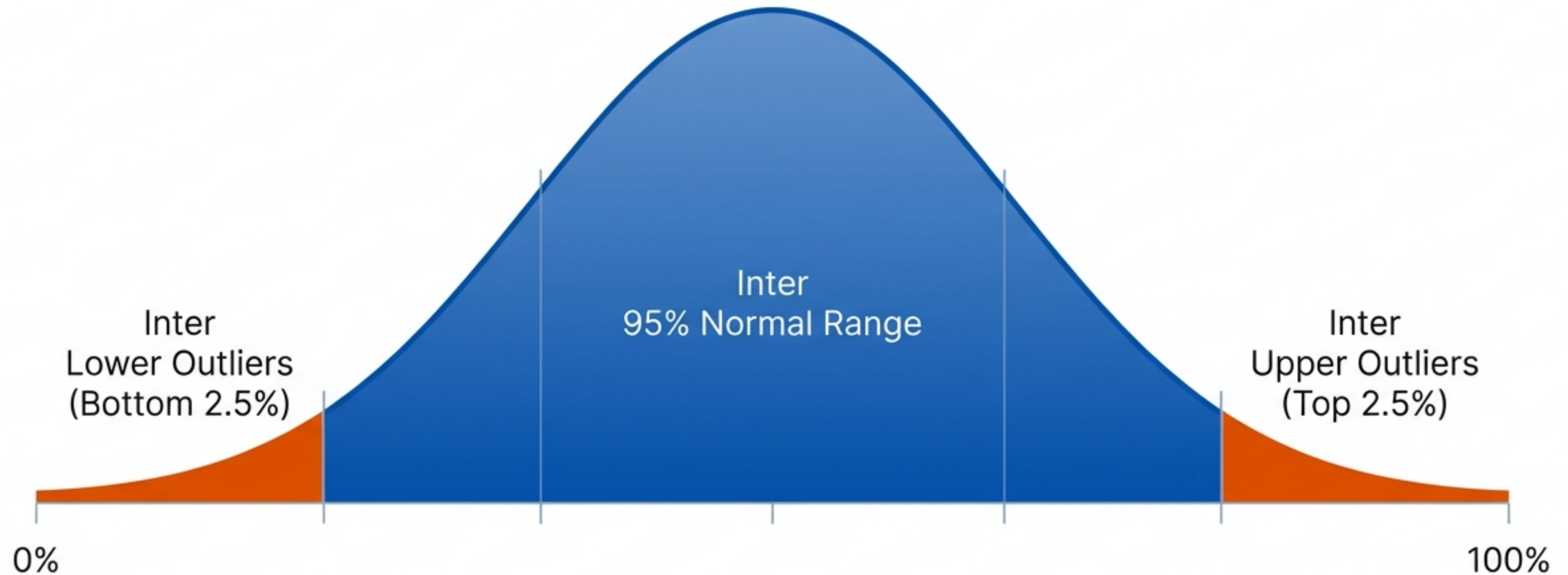


Taming the Extremes

Outlier Detection using the Percentile Method



A guide to refining datasets through Trimming and Winsorization

The Outlier Challenge

Outliers—data points significantly different from the majority—can skew analysis and break models. To handle them robustly, we need a method that respects the natural distribution of the data.

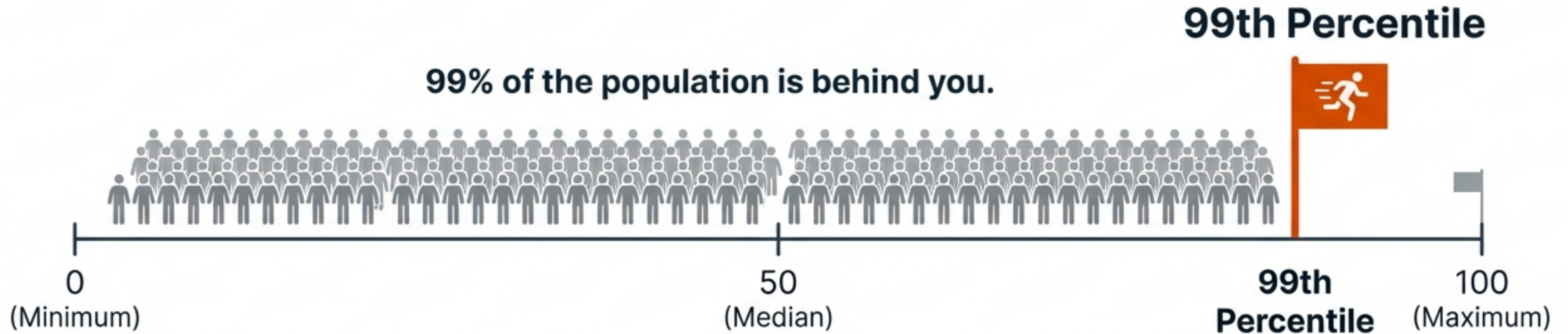
The Dataset context

We are analysing a dataframe containing 10,000 rows of human dimensions. Our primary focus is the 'Height' column.

	Gender	Height	Weight
	Male	73.84	241.89
	Male	68.78	162.31
	Male	74.11	212.74
	Female	58.91	102.01
	...	...	
10000	Female	61.23	115.34

Decoding Percentiles

The Exam Result Analogy



99th Percentile

Means you scored higher than 99% of people.

1st Percentile

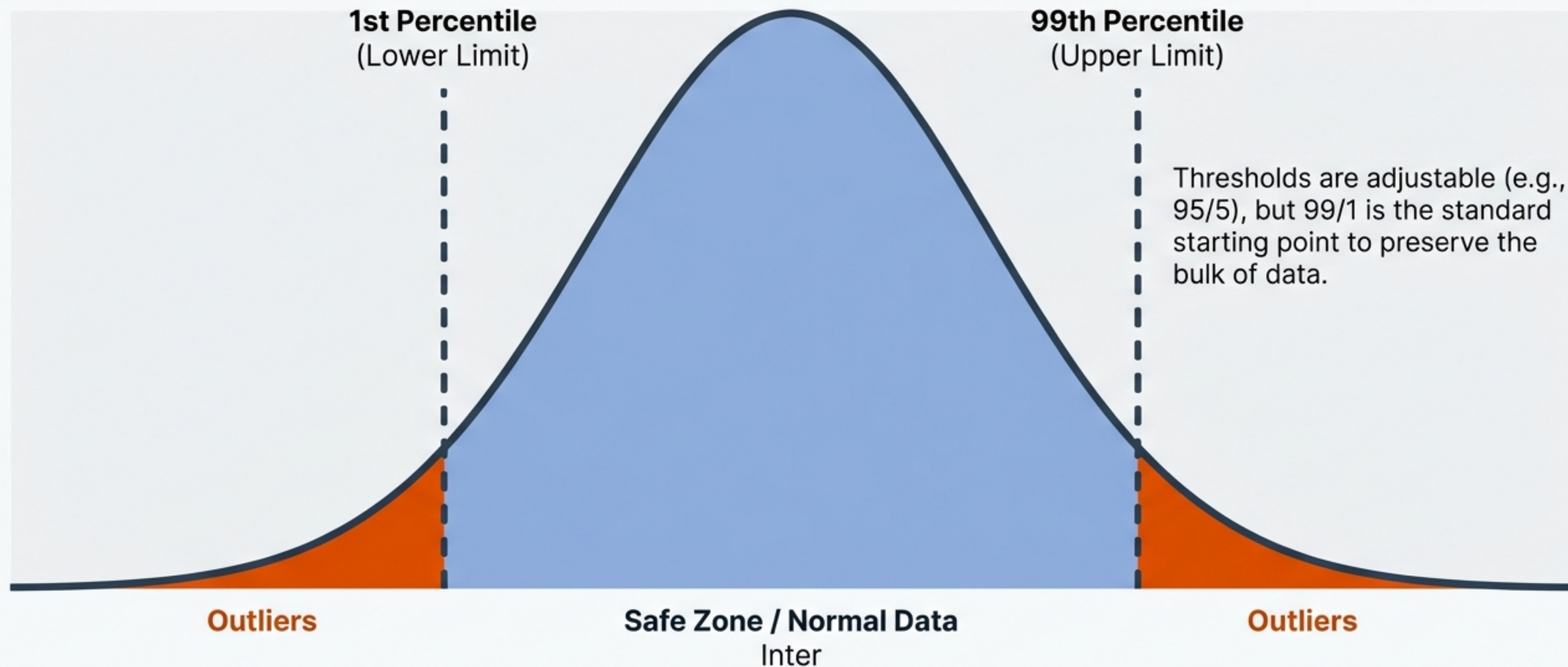
Means 99% of people scored higher than you.

50th Percentile

The Median. Half are ahead, half are behind.

Setting the Boundaries

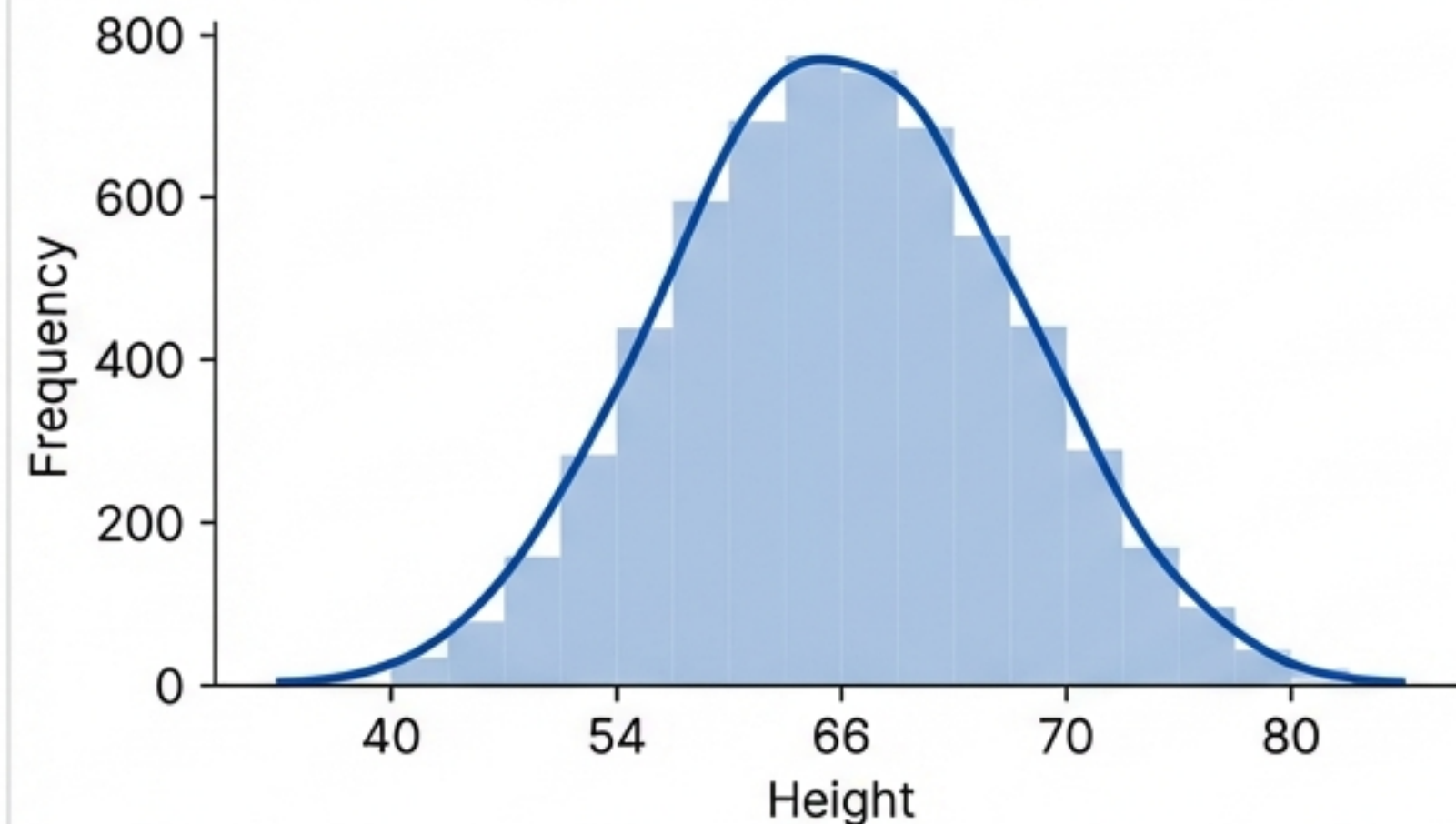
The Percentile Method identifies outliers by drawing lines in the sand to establish a “Safe Zone”.



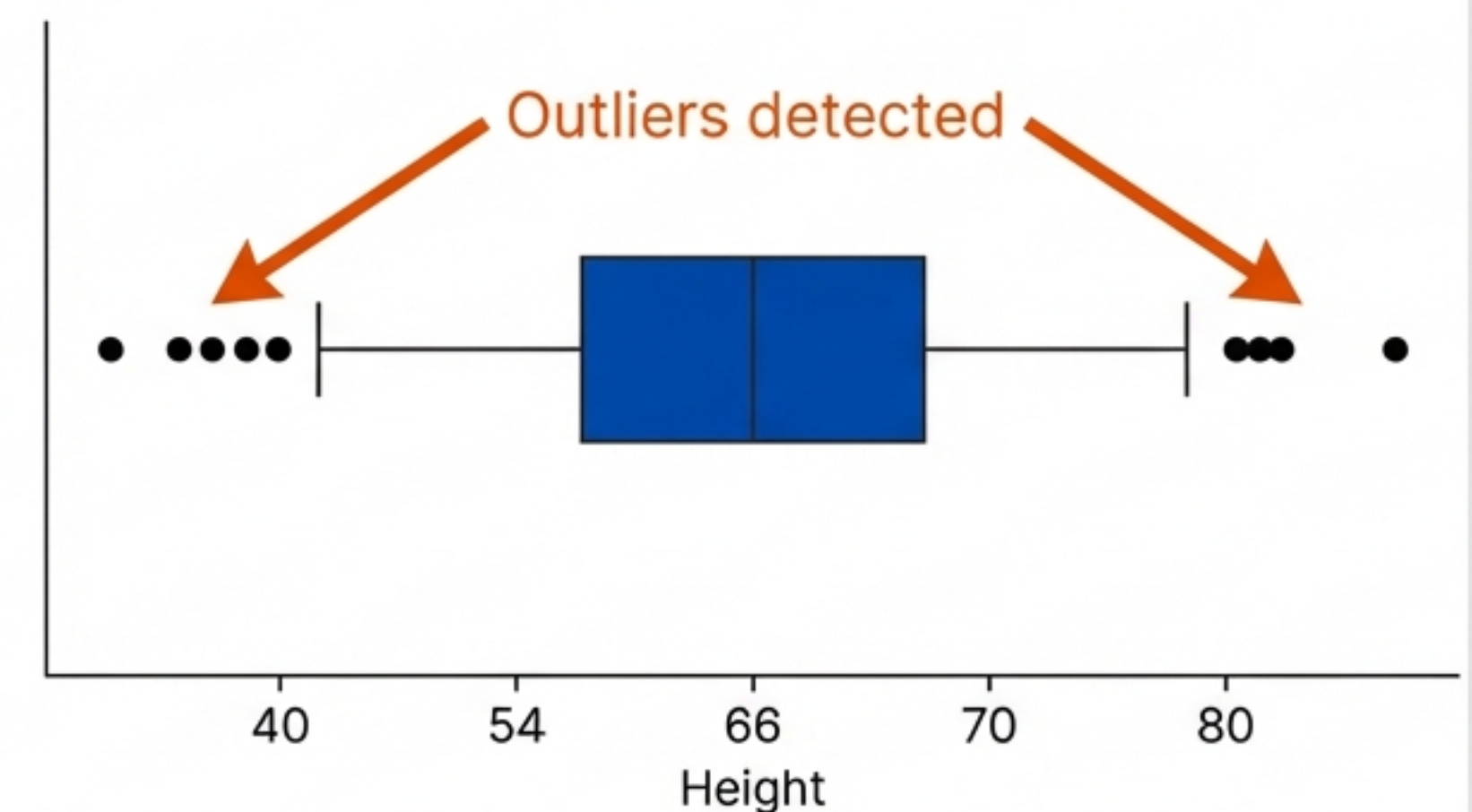
Step 1: Diagnosis

Inspecting the raw Height data

Distribution Plot (Seaborn distplot)



Boxplot (Seaborn boxplot)



Count: 10,000

Min: ~54.2

Mean: ~66.3

Max: ~78.9

Step 2: Calculating the Limits

We use the Pandas quantile() function to determine exact cut-off values.

```
upper_limit = df['Height'].quantile(0.99)  
lower_limit = df['Height'].quantile(0.01)
```

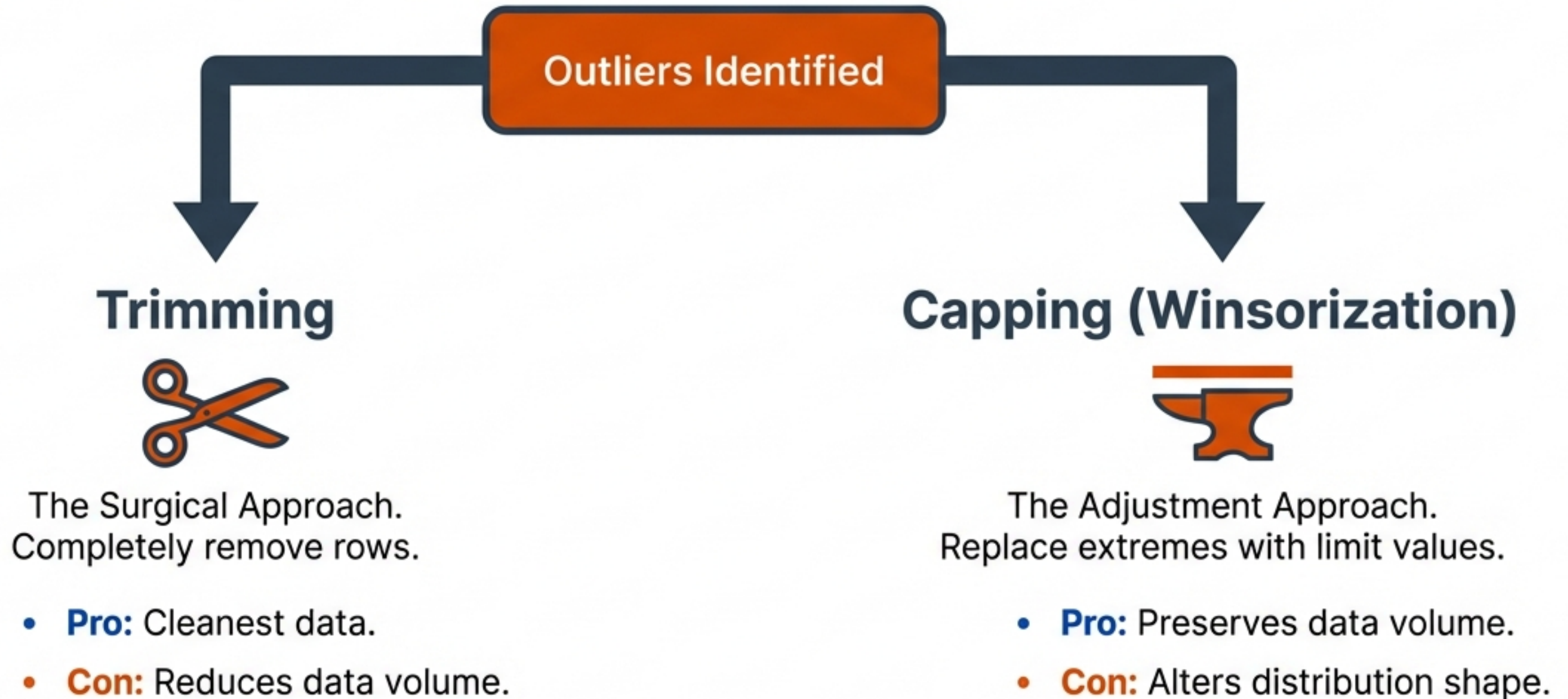
Upper Limit (0.99) : 74.78 inches

Lower Limit (0.01) : 58.13 inches

Logic defined: Any height above 74.78 or below 58.13 is treated as noise.

The Fork in the Road

Once outliers are identified, you have two strategic choices depending on your goals.



Path A: Trimming

The Surgical Approach

Trimming involves applying a Boolean mask to filter out the extremes. We keep only the rows where the height is within our calculated limits.

Intersection: Must satisfy both conditions.

```
new_df = df[(df['Height'] <= 74.78) & (df['Height'] >= 58.13)]
```

Index	Height	Data...
0	80.2	...
1	75.1	...
2	65.4	...
3	68.7	...
4	72.5	...
5	60.8	...
98	56.0	...
99	53.3	...

Values derived from quantiles.

Trimming Results

Verification

Status: Clean

Original Rows: 10,000

New Rows: ~9,800

Data Loss: ~200 rows removed

Boxplot of new_df['Height']

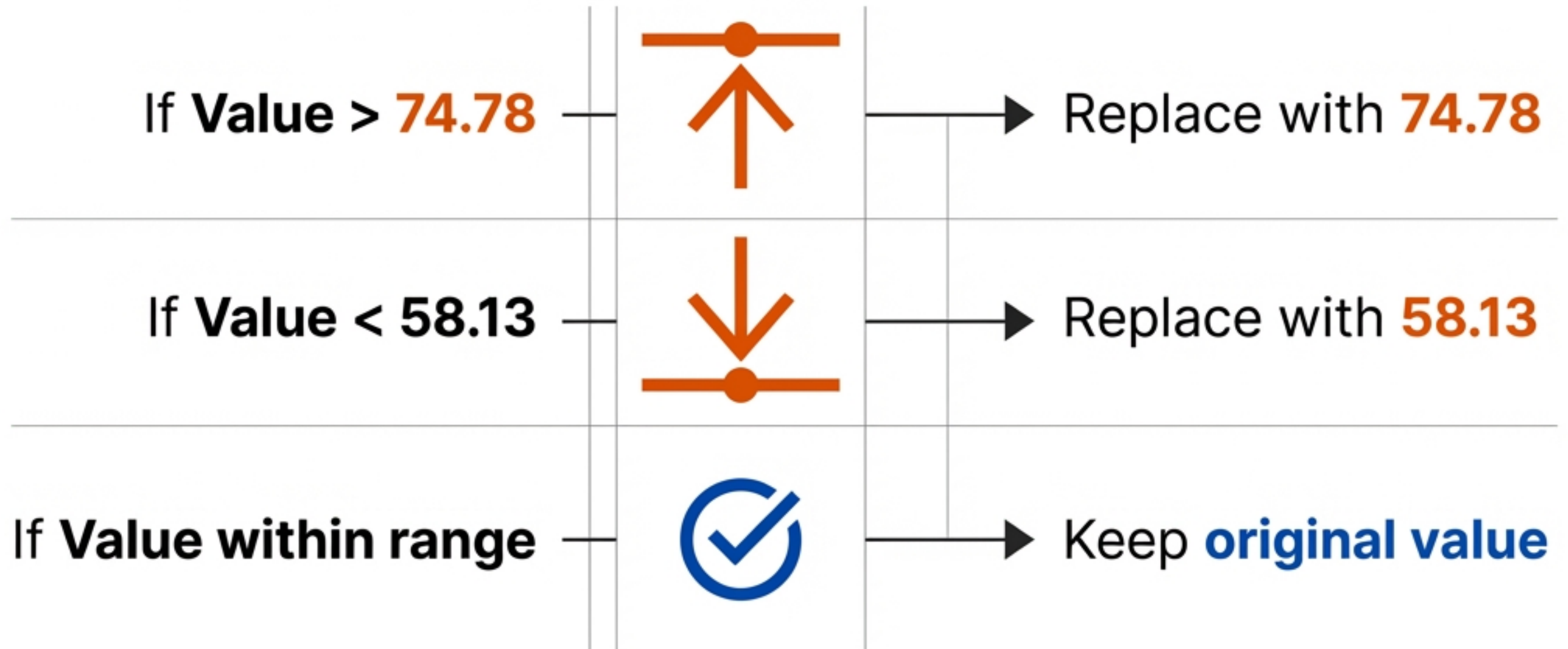


No outliers remaining.

Path B: Capping (Winsorization)

The Adjustment Approach

Named after statistician Charles Winsor, this technique caps data at the thresholds rather than deleting it. Useful when data volume is critical.



Implementing Capping

We use the `numpy.where` function to apply conditional logic efficiently.

```
# Apply Upper Limit
df['Height'] = np.where(df['Height'] >= upper_limit,
                        upper_limit,
                        df['Height'])

# Apply Lower Limit
df['Height'] = np.where(df['Height'] <= lower_limit,
                        lower_limit,
                        df['Height'])
```

Note:

While libraries like `scipy.stats` have built-in Winsorization functions, this manual method offers control without extra dependencies.

Capping Results

Verification

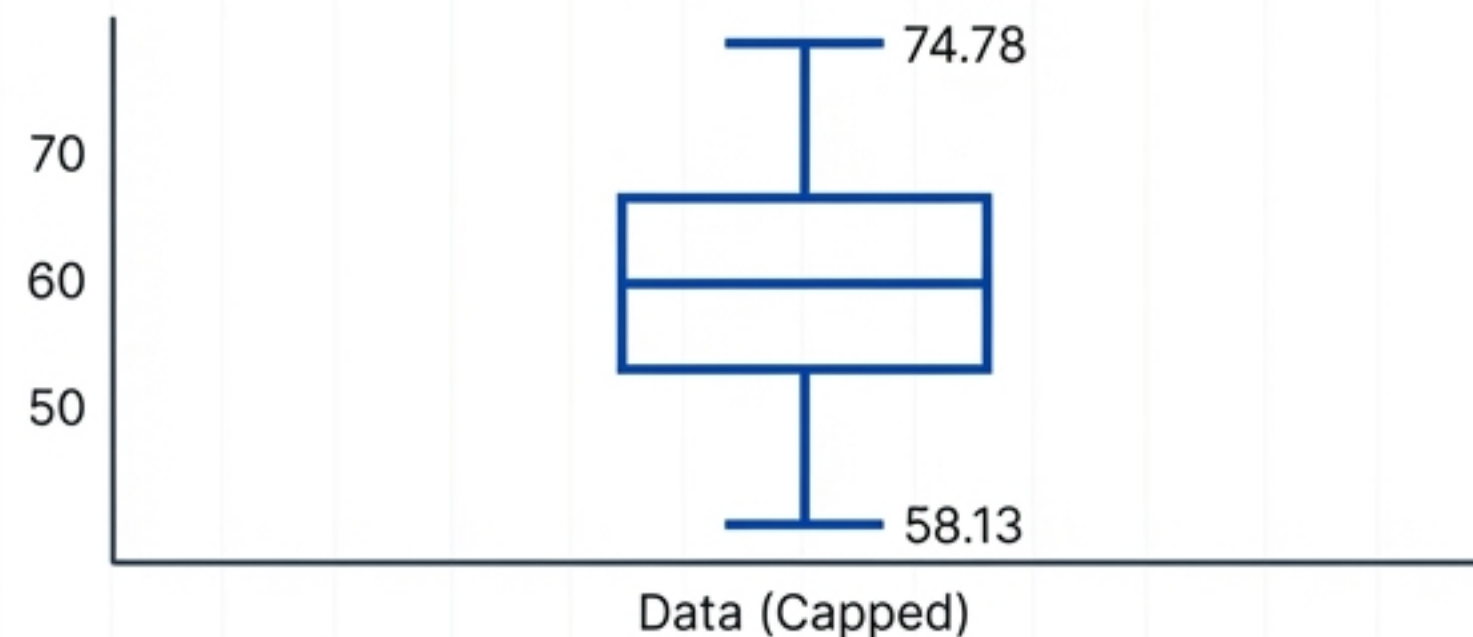
Inter Bold

Status: Capped ✓

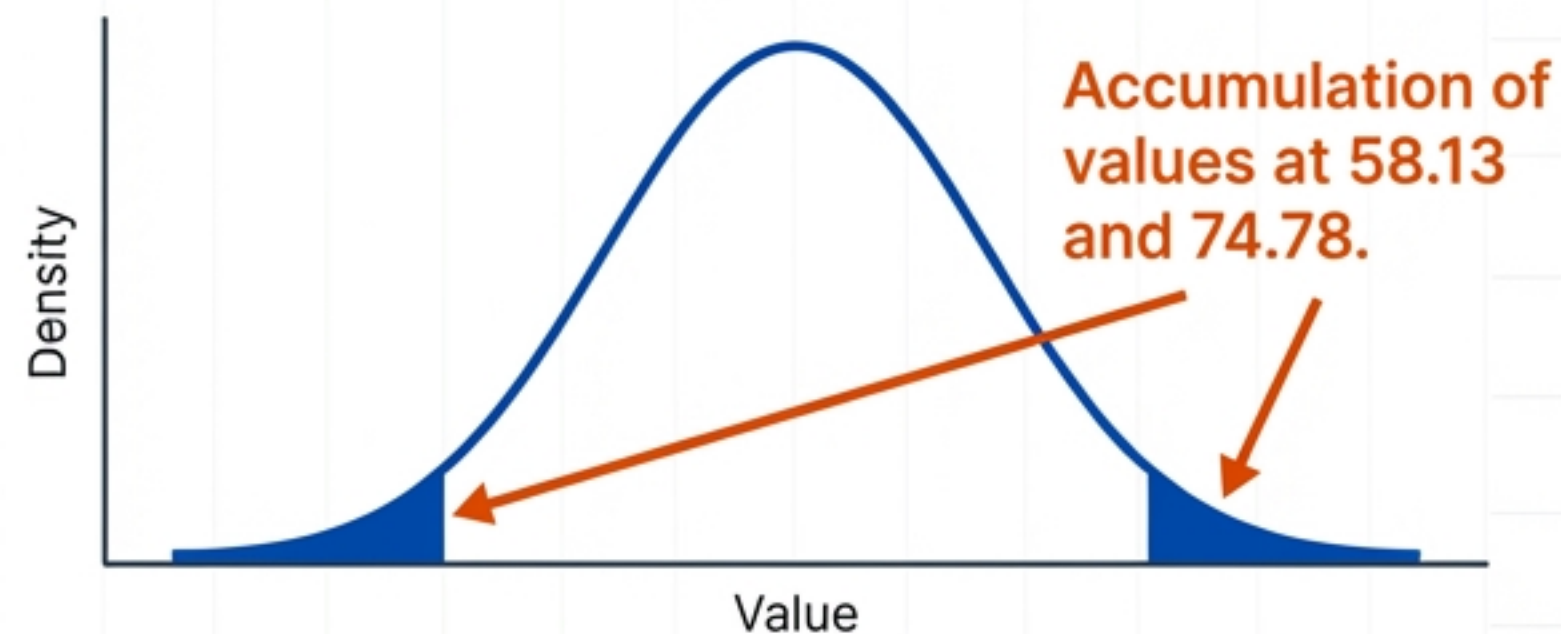
Rows: 10,000 (Preserved)

Impact: Outliers transformed to limit values.

Boxplot (After)

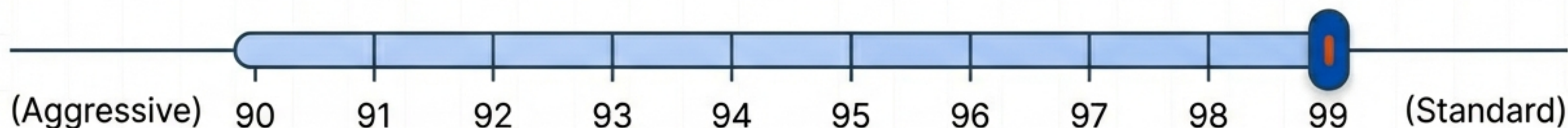


Distribution Note



Fine-Tuning the Method

The 99/1 split is a starting point, not a law. Thresholds should be tuned to your model's needs.



90th/10th (Aggressive):

Significant smoothing.
Alters data shape considerably.

95th/5th (Moderate):

Caps more data. Useful
for very noisy datasets.

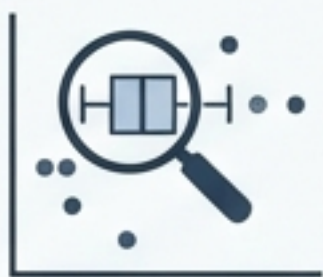
99th/1st (Standard):

Preserves most data.
Removes only extreme
noise.



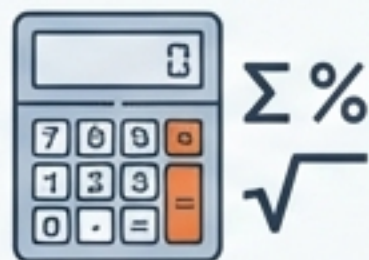
Tip: Compare model performance with different thresholds (e.g., 99.5 vs 99.0) to find the optimal balance.

Summary & Best Practices



Inspect

Use **Boxplots** to confirm outliers.



Calculate

Find 0.01 and 0.99 **quantiles**.



Choose

Trim (Remove) or **Cap** (Replace)?



Verify

Re-plot to ensure clean distribution.

Final Review: Method Comparison

Feature	Trimming	Capping (Winsorization)
Data Volume	Reduced (Loss of rows)	Preserved (Same row count)
Distribution Shape	Maintains original shape (narrower range)	Slightly alters shape (elevated edges)
Complexity	Simple Filtering	Conditional Replacement
Best For	Abundant data, extreme errors	Scarce data, preserving sample size

Choose the path that best aligns with your downstream modeling requirements.